

GAUTHAM BALACHANDRAN

gauthamgb85@gmail.com ◇ +91 63829 89533 ◇ GitHub ◇ LinkedIn

EDUCATION

Birla Institute of Technology and Science, Pilani

B.E. in Chemical Engineering | CGPA: 8.91/10

Pilani, Rajasthan

2023 – 2027

EXPERIENCE

Manav Robotics

Remote

Research Intern

Mar 2026 – Aug 2026

- Deployed vision-language-action (VLA) models and diffusion policies on a humanoid robot, taking learned manipulation policies from training to real-hardware execution.
- Built a reward-model-based filtering stage in the teleoperation data pipeline, automatically scoring and curating demonstrations to raise training-data quality for policy learning.
- Co-authored **IMBench**, a benchmark for intuitive manipulation (see Publications).

HiRo Lab, CPS Dept., IISc Bengaluru

Bengaluru, Karnataka

Research Intern

May 2025 – Jul 2025

- Implemented ODIL for bimanual imitation learning in MuJoCo with 3-stage visual servoing, achieving sub-centimeter placement precision. [Repo]
- Developed a dynamic throwing pipeline in Drake using analytical projectile modeling and optimization-based release control. [Repo]
- Benchmarked heuristic 3D bin-packing algorithms and co-designed a diffusion-based planner that outperformed them (see Publications).

Inspire Lab, BITS Pilani

Pilani, Rajasthan

Research Student

Jan 2025 – May 2026

- Developed scene-graph representations for structured environment understanding, modeling semantic and spatial relationships between objects for downstream multi-robot planning and navigation.
- Built segmentation-based mapping pipelines to extract object-level representations from visual observations and integrate them into persistent maps of dynamic environments.
- Integrated scene understanding with multi-robot coordination and navigation, enabling robots to reason about object relationships, changing environments, and task-relevant spatial information.

Criss Robotics

Pilani, Rajasthan

Software Subsystem Lead

Jan 2024 - Mar 2025

- Led the software subsystem to **3rd place** at the European Rover Challenge 2025 (Remote Category), owning system integration and mission execution.
- Engineered ROS pipelines for a mobile manipulator: 3D pose estimation via wheel-odometry/IMU/GPS fusion, plus autonomous navigation and 3D terrain mapping with A* and Dijkstra, validated in Unity simulation before field deployment.

PUBLICATIONS

IMBench: A Benchmark for Intuitive Manipulation | *CoRL 2026 SemRob Workshop*

- Proposed a benchmark and evaluation protocol for intuitive robot manipulation, enabling standardized comparison of VLA and diffusion-policy controllers across manipulation tasks.

DiffusionPack: Bin Packing with Custom Human Preferences | *NeurIPS 2025 Workshop (LAW)*

- Introduced a diffusion-based 3D bin-packing planner that ingests human preferences at test time via natural language through an LLM, achieving zero-shot adaptation to user-defined constraints with state-of-the-art packing efficiency.

TECHNICAL SKILLS

- **Languages:** Python, C/C++, SQL, JavaScript, C#
- **Robotics & ML:** ROS/ROS 2, MuJoCo, Drake, Unity, Imitation Learning, Diffusion Policies, VLAs, Computer Vision, Deep Learning
- **Libraries & Tools:** PyTorch, TensorFlow, NumPy, Pandas, Git/GitHub, Linux, VS Code